



Finishing

Created by	Gustavo Vieira
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Reviewers	Laurent Vincent Kishore Ganesan

1. Introduction

1.1 Purpose of the study

This document describes the life cycle inventory (LCI) model for fabric finishing operations as part of the Carbonfact Textile LCA Database. The aim is to provide a transparent, literature-based, and reproducible foreground inventory covering 18 globally representative finishing and pre-treatment datasets. Additional variants — cotton desizing-scouring-bleaching with IN and MX grid variants, and a China-specific polyester heat-setting dataset — were modelled during development as a sensitivity check on the global values and are not part of the published database. The model covers the main wet, thermal and mechanical finishing operations applied to woven and knitted fabrics in industrial textile production.

The model is designed to replace generic downstream processing proxies with process-specific LCIs that are independently verifiable.

1.2 Finishing processes — scope and classification

Fabric finishing encompasses all post-weaving and post-knitting operations that modify the physical, chemical, or aesthetic properties of the fabric. These operations are grouped into four categories for modelling purposes:

Table 1.2. Process classification and primary environmental drivers

Category	Datasets	Primary driver of environmental impact
Chemical finishing	Softening (Pad-Dry, Exhaust), DWR C6, DWR C0, Easy-care DMDHEU, Flame Retardant Pyrovatex,	Chemicals dominate for DWR C6 and FR processes; energy dominates for

Category	Datasets	Primary driver of environmental impact
	Flame Retardant Proban, Coating/Lamination	softening and coating
Thermal finishing	Heat Setting PES 190°C, Thermofixing Cotton 160°C, Sanforization, Can Dry, Calendering, Compacting	Thermal energy (natural gas, stenter or cylinder) is the dominant contributor in all cases
Mechanical finishing	Emerizing (peach skin), Raising/Brushing	Electricity only; no heat, no water, no chemicals
Fabric pre-treatment	Desizing (polyester, continuous), Mercerization	Chemicals (NaOH) dominate for mercerization; energy and water dominate for polyester desizing

1.3 System boundary

Gate-to-gate: energy, water and chemical inputs consumed during the finishing operation itself.

Included:

- Direct electricity (stenter drives, machine motors)
- Thermal energy (natural gas for stenter, steam for cylinder processes)
- Process water (pad bath, exhaust bath, washing)
- Chemical auxiliaries (softeners, DWR agents, FR agents, alkalis, bleaching agents, enzymes)
- Wastewater output
- Water vapour to air (Softening Pad-Dry only — from stenter evaporation)

Excluded:

- CED of chemical inputs (captured by ecoinvent background datasets)
- HVAC, lighting, and compressed air (not reported separately in BREF 2023; included in some All values)
- Upstream transport of fabric or chemicals
- Capital goods (machinery) — contribution to GWP is not significant relative to energy and chemicals
- Fabric mass losses (IOR = 1.0 for all processes; see Section 3.8)

1.4 Functional unit and scope

Parameter	Value
Functional unit	1 kg of finished dry fabric after the specified finishing operation (output-based, gate-to-gate)
Background	ecoinvent 3.12, system model: cut-off by classification

Parameter	Value
database	
Impact method	IPCC 2021 GWP100 (EF 3.1 midpoint characterisation factors) via Brightway2
Number of datasets	18 globally representative datasets (GLO geography)
IOR convention	IOR = 1.0 for all processes — finishing is a treatment, not a conversion; fabric substrate mass is conserved
Capital goods	Not modelled — contribution to GWP not significant relative to energy and chemicals
Dataset version	v3 (April 2026): incorporates All v0.9 values for sanforization, 4 new processes (Can Dry, Calendering, Compacting, Coating/Lamination)

2. Data sources and literature review

2.1 Review of existing LCI databases

ecoinvent 3.12 (cut-off) contains no dedicated textile finishing activities. The ecoinvent database provides background LCI data for chemicals, energy carriers, and water, but no foreground datasets for wet finishing, thermal finishing, or mechanical finishing operations. EF 3.1 is used exclusively as an impact characterisation method (characterisation factors), not as a source of foreground LCI data.

No alternative LCI database (SimaPro libraries, GaBi, or otherwise) covers the processes modelled here. Independent LCI development based on primary process literature is therefore necessary for all 18 datasets.

2.2 Primary literature sources

2.2.1 BREF Textiles 2023 - primary source for stenter-based processes

The BREF Textiles 2023 (JRC131874, Roth et al.) is the primary source for all stenter-based and exhaust-based processes. It provides multi-facility EU BAT (Best Available Technology) survey data covering electricity, thermal energy, water consumption, and wastewater generation. Key values used in this model:

Parameter	Value	Process applicability	Table / location
Electricity, stenter (gas-fired)	0.097 kWh/kg	All stenter-based processes (softening pad-dry, heat setting, thermofixing, DWR, easy-care, FR pad-dry)	Table 3.17; ESPR LCS2d
Thermal energy, stenter pad-dry	3.5 MJ/kg	Softening pad-dry, thermofixing, DWR, sanforization reference	BREF 2023 + ESPR LCS2d EU

Parameter	Value	Process applicability	Table / location
Thermal energy, heat setting PES	4.2 MJ/kg	Heat setting polyester at 190°C (higher temperature than cotton)	BREF 2023
Water, stenter pad-dry	0.0035 kg/kg	Processes using only stenter application without prior wetting	BREF 2023
Water, pre-dyeing DSB	60.0 kg/kg	Desizing-scouring-bleaching of woven cotton	BREF 2023
Electricity, pre-treatment	0.38 kWh/kg	Desizing-scouring-bleaching cotton (confirmed by Hasanbeigi & Price 2012)	BREF 2023 / LBNL-5744E
Thermal energy, pre-treatment	10.3 MJ/kg	Desizing-scouring-bleaching cotton	BREF 2023
Electricity, mercerization	0.20 kWh/kg	Continuous chain/chainless mercerizer	BREF 2023
Thermal energy, mercerization	2.0 MJ/kg	Mercerization (NaOH liquor heating)	BREF 2023

Cross-validation against the ESPR 3rd Preparatory Study (JRC, December 2025) confirms the stenter electricity value (0.097 kWh/kg) and the ecoinvent 3.12 activity choices. The ESPR model (sheet LCS2d, product "OtherWoven") uses identical energy values to BREF 2023.

2.2.2 BREF Textiles 2003 - secondary source for chemical dosages

The BREF Textiles 2003 is retained as a secondary source for chemical dosages where BREF 2023 does not report process-specific quantities (Section 4.5.1 for mercerization NaOH at 200 g/L; Table 4.20 for scouring and bleaching chemicals; Table 4.8 for softener owf ranges). The 2003 electricity values (0.3–0.5 kWh/kg stenter) are superseded by the 2023 value.

2.2.3 All v0.9 Facility Benchmark Calculator (November 2025)

The All/Made2Flow Facility Benchmark Calculator v0.9 (internal, November 2025) provides energy quantities for the four new v3 processes (Can Dry, Calendering, Compacting, Coating/Lamination) and a revised sanforization value. Key All v0.9 values adopted in this model:

Process (All name)	Total kWh/kg (All)	Split	Electricity (kWh/kg)	Heat (MJ/kg)
Fin-Sanforise	0.693	80% thermal / 20% electric	0.058	2.00
Fin-Emerising	0.200	100% electric	0.200	0
Fin-Brushing	0.200	100% electric	0.200	0
Fin-Can Dry	2.205	95% thermal / 5% electric	0.046	7.54

Process (All name)	Total kWh/kg (All)	Split	Electricity (kWh/kg)	Heat (MJ/kg)
Fin-Calendering	2.210	80% thermal / 20% electric	0.184	6.37
Fin-Compact	0.693	80% thermal / 20% electric	0.058	2.00
Fin-Coating/Lamination	2.400	80% thermal / 20% electric	0.200	6.91

⚠ Note on All emission factors: All v0.9 energy quantities are adopted; All GHG emission factors are NOT adopted. This model applies its own IPCC 2021-based EFs.

2.2.4 Hasanbeigi & Price (2012)

Hasanbeigi & Price (LBNL-5744E, 2012) provide multi-geography energy benchmarks for textile wet processing and finishing. Used as cross-check for pre-treatment (0.38 kWh/kg confirmed), emerizing (0.25 kWh/kg, superseded by All 0.20 kWh/kg in v3), and sanforization (0.15 kWh/kg, superseded by All with steam component in v3). Not used as primary source in v3 due to the availability of BREF 2023 and All v0.9.

2.2.5 Kamruzzaman (2016) - DMDHEU easy-care dosages

Kamruzzaman et al. (2016), published in the International Journal of Textile Science, provide process-specific dosages for DMDHEU easy-care finishing: resin 50 g/L, MgCl₂ catalyst 20 g/L, softener co-finish 1.25 g/L (Table 1). At 80% WPU these translate to 0.040 kg/kg DMDHEU, 0.016 kg/kg MgCl₂, and 0.001 kg/kg softener. These values are consistent with BREF 2003 ranges and are adopted as the primary source for DMDHEU chemical dosages.

2.2.6 NCBI Bookshelf (2011) and Lewis (2020) - FR dosages

The NCBI Bookshelf chapter on flame retardant textiles (2011) and Lewis et al. (2020) provide organophosphonate application rates used for both Pyrovatex (Clariant, N-methylol dimethylphosphonopropionamide) and Proban (Solvay, tetrakis(hydroxymethyl)phosphonium salt) processes. Pyrovatex: 120 g/L × 80% WPU = 0.096 kg/kg. Proban THPC: 150 g/L × 80% WPU = 0.120 kg/kg, plus 0.002 kg/kg urea co-reactant and 0.020 kg/kg H₂O₂ for the oxidation step.

2.2.7 ZDHC (2012) and Holmquist et al. (2016) - DWR dosages

The ZDHC Sustainable Research Report P05 (2012) and Holmquist et al. (2016, Environment International) provide DWR add-on rates for both C6 and C0 fluoropolymer and non-fluoro DWR agents. Both sources converge on <1% owf (0.8% owf used = 0.008 kg/kg) for the DWR agent. Holmquist et al. is the sole peer-reviewed LCA study available for DWR processes.

2.2.8 SGS ReTraze 2023 - China polyester facility

The SGS ReTraze third-party verification report (CN20231010001TAE, November 2023, v3.11) provides primary facility-level data for polyester desizing and polyester heat-setting. The report

covers Site 3, Product B (100% polyester woven fabric). The polyester desizing dataset is published with global (GLO) geography; the China-specific heat-setting PES variant was modelled during development as a sensitivity check and is not part of the published database. Cross-validation against BREF 2023 confirms the stenter electricity value (0.097 kWh/kg exact match); the higher thermal energy for PES heat-setting (8.75 MJ/kg vs. 3.5–4.2 MJ/kg EU) is expected given the higher processing temperatures required for polyester.

Parameter	BREF 2023 (EU BAT)	SGS ReTraze (CN facility)	Assessment
Stenter electricity	0.097 kWh/kg	0.097 kWh/kg (benchmark applied)	BREF benchmark
Heat, pre-treatment	10.3 MJ/kg (cotton)	9.05 MJ/kg (steam, PES)	PES requires less preparation than cotton
Heat, stenter finishing	3.5 MJ/kg (cotton)	8.75 MJ/kg (PES, 180–210°C)	PES heat-setting temperature substantially higher than cotton
Process water	60 kg/kg (cotton DSB)	37.5 kg/kg (PES desizing + finishing)	PES requires significantly less water than cotton

2.3 Chemical proxy selection

No ecoinvent 3.12 activity precisely matches the specialty chemicals used in textile finishing. Chemical proxies are selected based on structural analogy and GHG emission factor proximity. The table below summarises the rationale and uncertainty for each proxy.

Chemical function	Proxy used	ecoinvent activity	Rationale	Uncertainty
Aminosilicone softener	Silicone product	`market for silicone product \	RER`	Similar organosilicone chemistry and molecular weight range
C6 DWR fluorocarbon agent	PTFE/TFE polymer	`market for polytetrafluoroethylene \	GLO`	BREF 2023 recommended proxy for fluoropolymer DWR agents
C0 DWR non-fluoro agent	Silicone product	`market for silicone product \	RER`	Most C0 DWR agents are silicone- or wax-based; silicone is nearest proxy
DMDHEU resin	MUF adhesive (melamine urea formaldehyde)	`market for melamine urea formaldehyde adhesive \	GLO`	BREF 2023 recommendation; MUF shares the formaldehyde

Chemical function	Proxy used	ecoinvent activity	Rationale	Uncertainty
				crosslinking chemistry
DMDHEU catalyst (MgCl ₂)	MgCl ₂ from Ti sponge production	`market for magnesium chloride, from titanium sponge production \	GLO`	Only MgCl ₂ market available in ecoinvent 3.12; different production route
Pyrovatex CP (N-P organophosphonate)	Phosphoric acid, industrial	`market for phosphoric acid, industrial grade \	GLO`	Shares phosphorus backbone;
Proban THPC (organochlorophosphate)	Phosphoric acid, industrial	`market for phosphoric acid, industrial grade \	GLO`	Same proxy as Pyrovatex
Desizing/dispersing agents (PES)	Soap (surfactant)	`market for soap \	GLO`	Surfactant-based desizing agents; soap is nearest generic proxy

3. Life cycle inventory modelling

3.1 Energy model - the key structural distinction from spinning

Fabric finishing energy consumption is fundamentally different from spinning energy in one critical respect: finishing SEC is **mass-based and process-specific**, not a function of yarn fineness (dtex). This is because finishing operates on fabric, not on individual fibres, and the energy required per kilogram of fabric depends on the process type (drying temperature, bath ratio, mechanical load) rather than on the geometry of the constituent yarns.

Each finishing process has a single fixed SEC value (or at most a geography-specific variant for the same process) that is independent of fabric construction.

Energy functional form for all finishing processes:

$$SEC_{\text{electricity}} = k_{\text{elec}} \quad [\text{kWh/kg}]$$

$$SEC_{\text{heat}} = k_{\text{heat}} \quad [\text{MJ/kg}]$$

Where k_{elec} and k_{heat} are process-specific constants — no dtex argument, no material-type factor.

The stenter electricity constant of 0.097 kWh/kg is confirmed by three independent sources (BREF 2023, ESPR LCS2d, SGS ReTraze 2023) and applies to all stenter-based finishing processes regardless of fabric type or yarn count.

3.2 Stenter-based processes - energy parameterisation

All stenter-based processes share the same electricity value (0.097 kWh/kg) from BREF 2023. Thermal energy varies by process temperature and substrate:

Process	Electricity (kWh/kg)	Heat (MJ/kg)	Temperature	Source
Softening pad-dry (cotton/blends)	0.097	3.5	~130°C	BREF 2023
Thermofixing cotton	0.097	3.5	160°C	BREF 2023
Easy-care DMDHEU cure	0.097	4.0	160°C	BREF 2023
Heat setting PES (GLO)	0.097	4.2	190°C	BREF 2023
FR Pyrovatex cure	0.097	4.2	160°C	BREF 2023
Heat-setting PES stenter (CN)	0.097	8.75	180–210°C	SGS ReTraze 2023

The higher thermal value for CN heat-setting (8.75 MJ/kg vs. 4.2 MJ/kg) reflects the higher processing temperature range for polyester in the Chinese facility, as documented in the SGS ReTraze report. This value is adopted for the CN-specific dataset only; the GLO heat-setting value remains 4.2 MJ/kg based on BREF 2023.

3.3 Exhaust processes — energy parameterisation

Exhaust processes (jet dyeing machine, overflow, jigger) differ structurally from stenter processes: the fabric is processed in a bath at a defined liquor ratio, with heating of the bath liquid rather than hot-air drying.

Process	Electricity (kWh/kg)	Heat (MJ/kg)	Water (kg/kg)	Wastewater (m ³ /kg)
Softening Pad-Dry (stenter)	0.097	3.5	1.0	0.001
Softening Exhaust (jet/overflow)	0.12	1.5	20.0	0.018

The exhaust process has a higher electricity value (0.12 kWh/kg) due to the jet pump and agitation motor, but lower thermal energy (1.5 MJ/kg) because the bath is heated only once and the fabric does not require high-temperature drying. Water consumption is substantially higher (20 kg/kg at a typical 1:10 liquor ratio), and wastewater output is correspondingly higher.

3.4 Pre-treatment processes — energy and chemistry

Fabric pre-treatment (desizing, scouring, bleaching - collectively DSB) is the most energy- and water-intensive finishing process in this model. The GLO value from BREF 2023 is confirmed by Hasanbeigi & Price (2012) for electricity (0.38 kWh/kg) and represents a multi-step process sequence: enzymatic desizing → alkaline scouring → peroxide bleaching → washing.

3.5 Mechanical finishing — energy only

Emerizing (peach skin effect) and Raising/Brushing are purely mechanical processes. No heat, no water, no chemicals. Electricity only, from motor-driven abrasive drums or needle-roll cylinders. These have the lowest GHG impact of all finishing processes.

3.6 Chemical inputs — application rate derivations

Chemical add-on quantities (kg chemical per kg fabric) are derived from the pad application formula:

$$\text{Chemical add-on (kg/kg fabric)} = \frac{\text{Concentration [g/L]}}{1000} \times \text{WPU [fraction]}$$

Examples at standard WPU = 80%:

Chemical	Concentration	WPU	Add-on (kg/kg)	Source
Softener pad-dry (aminosilicone)	30 g/L	80%	0.030	BREF 2003 Table 4.8: 1–5% owf, central 3%
DMDHEU resin	50 g/L	80%	0.040	Kamruzzaman (2016) Table 1
MgCl ₂ catalyst	20 g/L	80%	0.016	Kamruzzaman (2016) Table 1
Pyrovatex CP	120 g/L	80%	0.096	NCBI 2011 Ch.4: 12% solids × 80% WPU
Proban THPC	150 g/L	80%	0.120	NCBI 2011 Ch.4: 15% solids × 80% WPU
NaOH mercerization	250 g/L	100%	0.250	BREF 2003 Section 4.5.1
C6/C0 DWR	0.8% owf	—	0.008	ZDHC 2012; Holmquist et al. 2016
Softener exhaust	5 g/L	—	0.015 (at 1:10 LR)	BREF 2003 / ZDHC 2012

3.7 Process water and wastewater

Process	Water input (kg/kg)	Wastewater (m ³ /kg)	Primary use
Softening Pad-Dry	1.0	0.001	Pad bath liquor; most evaporated in stenter
Softening Exhaust (jet)	20.0	0.018	Bath liquor at ratio 1:10 to 1:20
Sanforization	0.05	0	Steam humidification of rubber blanket
Mercerization	30.0	0.030	NaOH liquor bath and washing

Process	Water input (kg/kg)	Wastewater (m ³ /kg)	Primary use
DSB (cotton, GLO/IN/MX)	60.0	0.050	Desizing, scouring, bleaching baths + washing
DWR C6 / C0	0.0035	0.0035	Minimal pad bath water only
Easy-care DMDHEU	0.0035	0.001	Minimal pad bath water
FR Pyrovatex	0.0035	0.001	Minimal pad bath water
FR Proban THPC	30.0	0.030	Oxidation + washing steps after curing
Desizing Polyester (GLO)	35.0	0.035	Washing range for size removal
Heat-setting PES stenter (CN)	2.5	0.0002	Pad liquor only (functional finish bath)
Heat setting PES / Thermofixing / Emerizing / Raising / Can Dry / Calendering / Compacting / Coating	0	0	Dry processes — no water contact

3.8 IOR convention — the key difference from spinning models

$$\text{IOR} = \frac{1}{1 - L} \quad \text{where } L = 0 \Rightarrow \text{IOR} = 1 \text{ kg/kg for all finishing processes}$$

Rationale: Fabric substrate mass is conserved — finishing is a treatment, not a conversion. Chemical add-ons (softener 3% owf, FR 9.6–12% owf, DMDHEU 4–6% owf) increase fabric mass marginally, but these are modelled as chemical inputs, not fabric mass changes. The functional unit (1 kg finished dry fabric) is therefore equal to the input fabric mass (1 kg greige/dyed fabric).

Note: trace mass removal (emerizing removes fibre surface; mercerization may remove surface residues) is negligible and treated as IOR = 1.0 conservatively.

4. Life cycle impact assessment

4.1 LCIA results — GHG by process

Climate change impacts (kg CO₂eq / kg fabric) were computed using Brightway2 with IPCC 2021 GWP100 characterisation factors, ecoinvent 3.12 (cut-off) background, and the GLO average electricity mix unless otherwise noted.

Process	GHG energy (kgCO ₂ eq/kg)	Chemical GHG (kgCO ₂ eq/kg)	Dominant driver	Notes
Softening Pad-Dry	0.310	~0.255	Chemicals (silicone 0.03 kg × ~8.5)	Total ~0.57; silicone dominates over stenter heat

Process	GHG energy (kgCO ₂ eq/kg)	Chemical GHG (kgCO ₂ eq/kg)	Dominant driver	Notes
Softening Exhaust (Jet)	0.187	~0.128	Energy (heat 1.5 MJ)	Total ~0.31; lower total than pad-dry
Heat Setting PES 190°C	0.358	0	Energy (heat 4.2 MJ)	Pure thermal; no chemicals
Thermofixing Cotton 160°C	0.310	0	Energy (heat 3.5 MJ)	Same stenter profile as softening but no chemicals
Sanforization	0.179	0	Energy (heat 2.0 MJ)	v3 adds steam; previous v2 had only electricity
Desizing Polyester (GLO)	—	—	Energy + wastewater (heat 9.0 MJ, elec 0.46 kWh, 35 kg water)	Total 1.29 kgCO ₂ eq/kg (published); polyester size removal + washing
Mercerization	0.276	~0.21	Chemicals (NaOH 0.25 kg)	Total ~0.49; NaOH ~1.0 kgCO ₂ eq/kg
DWR C6	0.310	~0.20	Chemicals (TFE proxy 0.008 kg × ~25)	Total ~0.51; C6 proxy dominates (±50% uncertainty)
DWR C0	0.310	~0.068	Energy	Total ~0.38; silicone proxy 0.008 kg × ~8.5
Easy-care DMDHEU	0.344	~0.16	Energy + chemicals roughly equal	MUF resin 0.04 kg × ~4 kgCO ₂ eq/kg
Emerizing	0.138	0	Energy (electricity 0.2 kWh)	Lowest GHG of all processes
Raising/Brushing	0.138	0	Energy (electricity 0.2 kWh)	Equal to emerizing
FR Pyrovatex	0.358	~0.014 (proxy — underestimate 3–8×)	Energy (chemical severely underestimated)	True total could be 0.37–1.13 kgCO ₂ eq/kg
FR Proban THPC	0.587	~0.018 (proxy — underestimate 3–8×)	Energy (high elec 0.5 kWh + heat)	True total could be 0.59–1.55 kgCO ₂ eq/kg
Can Dry	0.555	0	Energy (heat 7.54 MJ)	High thermal — cylinder contact drying
Calendering	0.569	0	Energy (heat 6.37 MJ)	High thermal
Compacting	0.179	0	Energy (heat 2.0 MJ)	Same profile as sanforization

Process	GHG energy (kgCO ₂ eq/kg)	Chemical GHG (kgCO ₂ eq/kg)	Dominant driver	Notes
Coating/Lamination	0.617	0	Energy (heat 6.91 MJ)	Highest thermal of the 4 new v3 processes

4.2 Comparison with EF 3.1

EF 3.1 does not contain dedicated fabric finishing datasets. The comparison is therefore not dataset-to-dataset but rather a check of energy and chemical GHG contributions against the EF 3.1 characterisation factor framework. The use of IPCC 2021 GWP100 characterisation factors (as implemented in EF 3.1 via Brightway2) is consistent with this framework. No systematic differences are expected at the characterisation level; all discrepancies relative to any existing EF 3.1-based finishing estimates will derive from differences in foreground LCI values, not from characterisation.

5. Inferences and limitations

Four categories of limitations are relevant to users of these datasets.

First, **chemical proxy quality** is the dominant source of model uncertainty for chemical-intensive processes. The phosphoric acid proxy for FR agents (Pyrovatex and Proban) underestimates the true GHG contribution of these chemicals by a factor of 3–8. The TFE/PTFE proxy for C6 DWR agents carries ±50% uncertainty. These processes should not be used for LCA comparisons involving FR or C6 DWR treatments without acknowledging this systematic underestimate.

Second, **All v0.9 data quality** is limited for the four new v3 processes (Can Dry, Calendaring, Compacting, Coating/Lamination). The All dataset is an internal benchmark tool, not a peer-reviewed source. Energy values for these processes carry higher uncertainty (±30–40%) than BREF-based values (±15–20%). Calibration against facility metering data for these processes is a priority for v4.

Third, **geographic coverage** is limited. GLO values are derived primarily from EU BREF data, which may not represent production conditions in major manufacturing countries (Bangladesh, Vietnam, Turkey). The three India, Mexico, and China variants address a fraction of global finishing capacity. Where geography is unknown, the GLO value is the appropriate default, with the understanding that it may over- or under-estimate process GHG by 10–30%.

Fourth, **HVAC, compressed air, and lighting** are excluded from all datasets. BREF 2023 reports electricity values as machine-specific (stenter drives, pumps, controls) without ancillary loads. This is consistent across all stenter-based datasets but may understate total facility electricity by 10–20% compared with facility-level metering.

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7. Annex

Annex A — ecoinvent 3.12 activity mapping (verified UUIDs)

Exchange	ecoinvent activity name	Geography	Unit	UUID
Electricity (GLO)	market group for electricity, medium voltage	GLO	kWh	1d8b7893
Electricity (IN)	market group for electricity, medium voltage	IN	kWh	b40ba24c
Electricity (MX)	market for electricity, medium voltage	MX	kWh	003bd44d
Heat (natural gas)	market for heat, district or industrial, natural gas	RoW	MJ	6bc6e9df
Tap water	market for tap water	RoW	kg	ac46debc
Acetic acid	market for acetic acid	GLO	kg	c20547f0
Sodium hydroxide 50%	market for sodium hydroxide, without water, in 50% solution state	RER	kg	42daa75e
Hydrogen peroxide 50%	market for hydrogen peroxide, without water, in 50% solution state	RER	kg	ddc30e4a
Silicone product (softener/DWR CO proxy)	market for silicone product	RER	kg	dbc40d7d
Enzymes (alpha-amylase proxy)	market for enzymes	GLO	kg	5950293a
Soap (surfactant proxy)	market for soap	GLO	kg	b113a26b
MUF adhesive (DMDHEU proxy)	market for melamine urea formaldehyde adhesive	GLO	kg	9f1a32c4
MgCl ₂ (catalyst proxy)	market for magnesium chloride, from titanium sponge production	GLO	kg	8ca462c0
Urea	market for urea	RoW	kg	7f09574a
Sulfuric acid	market for sulfuric acid	RER	kg	ff59f020
PTFE/TFE (C6 DWR proxy)	market for polytetrafluoroethylene	GLO	kg	bddf47ba
Phosphoric acid (FR proxy — HIGH PRIORITY)	market for phosphoric acid, industrial grade	GLO	kg	21472a4d
Wastewater, textile	market for wastewater from textile production	RoW	m ³	9f14ddec

Annex B — LCI for 1 kg finished fabric — Softening, Pad-Dry, Stenter (representative example)

Activity	Value	Unit	Source / Rationale
Electricity (input)	0.097	kWh	Stenter pump/control/drive electricity. BREF 2023 (Roth et al.) via ESPR LCS2d EU. `market group for electricity, medium voltage \
Heat, natural gas (input)	3.5	MJ	Stenter hot-air oven at ~130°C. BREF 2023. `market for heat, district or industrial, natural gas \
Tap water (input)	1.0	kg	Pad bath liquor. BREF 2023. `market for tap water \
Silicone product / aminosilicone softener (input)	0.030	kg	30 g/L × 80% WPU = 0.030 kg/kg. BREF 2003 Table 4.8: 1–5% owf, central 3%. [PROXY] `market for silicone product \
Acetic acid / pH adjustment (input)	0.001	kg	pH adjustment of pad bath. BREF 2023 ESPR. [ESTIMATE][SINGLE SOURCE]. `market for acetic acid \
Water vapour to air (output)	0.8	kg	Evaporated in stenter dryer. Mass balance: 1.0 kg water in – 0.001 m³ WW = ~0.8 kg evaporated. [ESTIMATE]. Biosphere elementary flow.
Wastewater, textile (output)	0.001	m³	Pad bath overflow and preparation losses. BREF 2023. `market for wastewater from textile production \
Fabric substrate (IOR note)	1.000	kg	IOR = 1.0. No fabric mass lost. 1 kg fabric in = 1 kg finished fabric out. No IOR caller convention needed.

Annex C — LCI for 1 kg finished fabric — Desizing-Scouring-Bleaching, Woven Cotton, GLO

Activity	Value	Unit	Source / Rationale	
Electricity (input)	0.38	kWh	Hasanbeigi & Price 2012 (LBNL-5744E), confirmed by BREF 2023. `market group for electricity, medium voltage \	GLO` UUID: 1d8b7893.
Heat, natural gas (input)	10.3	MJ	BREF 2023. Heating of desizing, scouring, and bleaching baths; highest thermal load of all datasets. `market for heat, district or industrial, natural gas \	RoW` UUID: 6bc6e9df.

Activity	Value	Unit	Source / Rationale	
Tap water (input)	60.0	kg	BREF 2023: 60 kg/kg for cotton DSB. `market for tap water \	RoW` UUID: ac46debc.
Enzymes, alpha-amylase (input)	0.003	kg	Desizing step: 0.3 g/L × 10 L/kg liquor ratio = 0.003 kg/kg. [SINGLE SOURCE: Udhayamarthandan & Srinivasan 2019]. `market for enzymes \	GLO` UUID: 5950293a.
Sodium hydroxide 50% — scouring (input)	0.015	kg	BREF 2003 Table 4.20, p.166. `market for sodium hydroxide, without water, in 50% solution state \	RER` UUID: 42daa75e.
Soap / wetting agent — scouring (input)	0.005	kg	Surfactant for scouring. BREF 2003 Table 4.20. `market for soap \	GLO` UUID: b113a26b.
Hydrogen peroxide 50% — bleaching (input)	0.025	kg	H ₂ O ₂ bleaching agent. BREF 2003 Table 4.20. `market for hydrogen peroxide, without water, in 50% solution state \	RER` UUID: ddc30e4a.
Sodium hydroxide 50% — bleaching (input)	0.005	kg	NaOH for alkaline H ₂ O ₂ bleaching. BREF 2003 Table 4.20. `market for sodium hydroxide, without water, in 50% solution state \	RER` UUID: 42daa75e.
Wastewater, textile (output)	0.050	m ³	Mass balance: 60 kg water input = 0.05 m ³ wastewater output. `market for wastewater from textile production \	RoW` UUID: 9f14ddec.

Annex D — LCI for 1 kg finished fabric — DWR C6, Fluorocarbon, Pad-Dry, Stenter

Activity	Value	Unit	Source / Rationale	
Electricity (input)	0.097	kWh	BREF 2023. Same stenter benchmark. `market group for electricity, medium voltage \	GLO` UUID: 1d8b7893.
Heat, natural gas (input)	3.5	MJ	BREF 2023. Same stenter profile as softening. `market for heat, district or industrial, natural gas \	RoW` UUID: 6bc6e9df.
Tap water (input)	0.0035	kg	Minimal pad bath (DWR applied as very dilute solution). BREF 2023. `market for tap water \	RoW` UUID: ac46debc.

Activity	Value	Unit	Source / Rationale	
Fluorinated polymer, C6 TFE proxy (input)	0.008	kg	C6 DWR add-on: 0.8% owf = 0.008 kg/kg. ZDHC 2012 (<1% owf); Holmquist et al. 2016 (confirmed). [PROXY ±50% uncertainty. DOMINANT GHG CONTRIBUTOR: 0.008 kg × ~25 kgCO ₂ eq/kg = ~0.20 kgCO ₂ eq/kg]. `market for polytetrafluoroethylene \	GLO` UUID: bddf47ba.
Acetic acid / pH adjustment (input)	0.001	kg	[ESTIMATE][SINGLE SOURCE]. `market for acetic acid \	GLO` UUID: c20547f0.
Wastewater, textile (output)	0.0035	m ³	Mass balance. `market for wastewater from textile production \	RoW` UUID: 9f14ddec.

Annex E — Complete energy and water summary — all 18 published datasets

Dataset	Elec (kWh/kg)	Heat (MJ/kg)	Water (kg/kg)	WW (m ³ /kg)	Source tier
Softening Pad-Dry	0.097	3.5	1.0	0.001	Tier 2 (BREF 2023)
Softening Exhaust (Jet)	0.12	1.5	20.0	0.018	Tier 2 (BREF 2023)
Heat Setting PES 190°C	0.097	4.2	0	0	Tier 2 (BREF 2023)
Thermofixing Cotton 160°C	0.097	3.5	0	0	Tier 2 (BREF 2023)
Sanforization	0.058	2.0	0.05	0	Tier 3 (All v0.9)
Desizing Polyester	0.46	9.0	35.0	0.035	Tier 3 (SGS ReTraze 2023)
Mercerization	0.20	2.0	30.0	0.030	Tier 2 (BREF 2023)
DWR C6	0.097	3.5	0.0035	0.0035	Tier 2 (BREF 2023)
DWR C0	0.097	3.5	0.0035	0.0035	Tier 2 (BREF 2023)
Easy-care DMDHEU	0.097	4.0	0.0035	0.001	Tier 2 (BREF 2023 / Kamruzzaman 2016)
Emerizing	0.20	0	0	0	Tier 3 (All v0.9)

Dataset	Elec (kWh/kg)	Heat (MJ/kg)	Water (kg/kg)	WW (m ³ /kg)	Source tier
Raising/Brushing	0.20	0	0	0	Tier 3 (All v0.9)
FR Pyrovatex	0.097	4.2	0.0035	0.001	Tier 2 (BREF 2023 / NCBI 2011)
FR Proban THPC	0.50	3.5	30.0	0.030	Tier 2 (BREF 2023 / Lewis 2020)
Can Dry	0.046	7.54	0	0	Tier 3 (All v0.9)
Calendering	0.184	6.37	0	0	Tier 3 (All v0.9)
Compacting	0.058	2.0	0	0	Tier 3 (All v0.9)
Coating/Lamination	0.200	6.91	0	0	Tier 3 (All v0.9)

(Development/sensitivity variants — modelled but not published: cotton Desizing-Scouring-Bleaching GLO/IN/MX: 0.38 kWh/kg, 10.3 MJ/kg, 60 kg/kg water; Heat-Setting PES CN: 0.097 kWh/kg, 8.75 MJ/kg, 2.5 kg/kg water.)

Annex F — Blueprint default status for product LCA integration

Dataset	Blueprint default	Applicable fabric types
Softening Pad-Dry	UNIVERSAL DEFAULT — activate for all wovens post wet-processing	All fibres; all wovens and most knits processed in stenter
Softening Exhaust (Jet)	DEFAULT for knits and delicate substrates	Knits, wool, silk, fine synthetics
Heat Setting PES 190°C	MANDATORY DEFAULT for any fabric with synthetic fibre content	Any fabric with >20% polyester, polyamide, or elastane
Thermofixing Cotton 160°C	DEFAULT for cotton/MMCF after any pad-dry chemical finishing	Cotton and MMCF wovens/knits after softening, DWR, or easy-care application
Sanforization	DEFAULT for cotton wovens	Cotton wovens (shirting, denim, poplin, twill)
Pre-dyeing DSB (GLO/IN/MX)	MANDATORY DEFAULT before any cotton woven dyeing	Greige woven cotton and cellulosic wovens
Compacting	DEFAULT for cotton jersey/interlock knits	Cotton jersey, interlock, single-knit, double-knit
Mercerization	OPTIONAL — premium cotton only	Premium shirting, high thread-count bedding
DWR C6 / C0	CONDITIONAL — outerwear category only	Outerwear shells, performance outdoor, waterproof categories
Easy-care DMDHEU	CONDITIONAL — workwear, non-iron shirting	Formal shirts, workwear, uniforms
Emerizing	NICHE — microfibre products only	Polyester/nylon microfibre knits

Dataset	Blueprint default	Applicable fabric types
Raising/Brushing	NICHE — fleece/flannel products only	Flannel, fleece, brushed sweatshirt fabrics
FR Pyrovatex / FR Proban	INDUSTRIAL ONLY — never a general default	PPE, oil & gas workwear, electrical utility, defence
Can Dry / Calendering / Coating	CONDITIONAL — specific product requirements	Can Dry: terry/tubular knits; Calendering: high-gloss; Coating: technical outerwear